Akash Chowdary Kodali

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AI Engineer | LLM systems, RAG, voice agents and evaluation

Available Jan 2027 | F-1 OPT, STEM OPT eligible | Open to relocation

Summary

M.S. CS student at California State University, Long Beach (CSULB), graduating in December 2026, with a first-author ACL Anthology paper (WASP workshop, IJCNLP-AACL 2025) on how often LLMs get citations right. At CSULB's IT Services I build AI systems, including a voice agent deployed to production and in staged business testing, and a multichannel commencement assistant.

Skills

AI & LLM: OpenAI Whisper, Azure Voice Live, tool calling, RAG, Azure AI Search, OpenAI and Gemini APIs, prompt-injection defense, LoRA, llama.cpp, PyTorch
Evaluation: paired bootstrap, McNemar, Cohen's kappa, LLM judge, roofline
Engineering: Python, FastAPI, TypeScript, React, Azure Container Apps, App Insights, Service Bus, pytest

Experience

Software Engineer (Student Assistant) Dec 2025 - Present
Innovation Team (ITS), California State University, Long Beach

- Designed and built an AI commencement assistant for the 2027 commencement: 4 services on Azure Container Apps joined by Service Bus session queues, answering from a 19-topic knowledge base on gpt-4o-mini across 6 channels built, 4 enabled (SMS, web, Teams, Copilot).
- Fixed the production DocuSign Connect router on Azure Functions that parses completed forms (CRM accounts, graduation, course completion, alumni events) and routes them to Logic Apps, Power Automate or microservices; fixed 4 filename variants it was silently dropping.
- Sole engineer on the Alumni Event Planner, in final testing on Azure ahead of launch for CSULB alumni events: React 19 and TypeScript front end, FastAPI on Azure Functions, Entra ID roles and offline-tolerant QR check-in, about 1,250 backend pytest and 120 frontend tests.
- Led the design of the Student Financial Services voice agent and extended a Microsoft sample into production (117 commits): bilingual calls via OpenAI Whisper and Azure Voice Live (gpt-realtime), in staged business testing before students use it.
- Hardened the voice agent for production: fixed a response-lifecycle race that announced a transfer 3 times without ever attempting it (run-once transfer plus a 6-second watchdog), and added WebSocket authentication, webhook secret validation, a path-traversal guard and a readiness probe.
- Built the Azure AI Search RAG layer (hybrid BM25 + vector + semantic ranking), raised per-result context from 800 to 2,000 characters after truncation dropped key facts, and cut the system prompt 35% to answer only from search results.
- Traced **34 failed call transfers** in 30 days (87% with one platform error code) to a missing caller ID on the outbound phone leg using App Insights logs; shipped a fix that passes an owned caller ID with a retry-then-graceful-hangup path, and moved transfer and end-call from fragile text markers to structured function tools.
- Added a deterministic SMS opt-out compliance gate (STOP never reaches the LLM), prompt-injection defense with 9 checks, a 14-URL link allow-list and per-conversation rate limits, covered by 107 tests.

Graduate Research Assistant Jan 2025 - Dec 2026
California State University, Long Beach

- M.S. thesis, Atlas: Prompting, Web Search and Verification in Large Language Model Bibliographic Retrieval. Measured how prompting, web search (OpenAI and Gemini APIs with tool calling) and verification change the DOI accuracy of GPT and Gemini against Crossref's production matcher on its 2,000-reference benchmark; Gemini with web search matched the accuracy of Crossref's purpose-built production matcher (**2.6 F1** gap, not statistically significant).
- Designed the evaluation: paired bootstrap (5,000 resamples) with adjusted p-values, McNemar tests, blind dual human annotation with hidden controls plus an LLM judge (agreement by Cohen's kappa), and a rule-based verifier; wrote 90 of 109 commits in the lab's Python repo.
- Built the experiment platform for a multi-agent study of coordination loss when a teammate is swapped mid-task (Overcooked-AI, 450 episodes, Value-at-Risk scoring), including an LLM agent layer for OpenAI, Gemini and Ollama.

Projects

Cross-Platform LLM Inference Benchmark *Python, llama.cpp, PyTorch* Spring 2026

- Profiled LLaMA 3.2 1B/3B decoding in llama.cpp on Apple M4, M4 Pro, NVIDIA A100 and x86 across F16, Q8 and Q4 with PyTorch hooks and a roofline model: decode throughput tracks memory bandwidth ($R^2 > 0.99$), and 4-bit quantization gives **2.37x** on M4 but only 1.07x on A100.

Ontology-Grounded Vision-LLM Risk Detection *Python, OWL/RDF, rdflib, GPT-4o-mini, Gemini, Pixtral* Spring 2026

- Grounded three vision LLMs in a custom driving-risk ontology across 972 calls: valid structured (RDF) output rose from **60.1% to 96.9%** and precision rose 45% (relative), at a 41% recall cost.

Health Insurance Q&A Agent *FAISS, LoRA, Gemma-2-2B, FastAPI, Streamlit* Fall 2025

- Built FAISS retrieval over 6,021 CMS Marketplace documents, fine-tuned Gemma-2-2B with LoRA on 18,020 synthetic Q&A pairs, and routed live-data questions to the healthcare.gov API, served through FastAPI and Streamlit.

Publications

Kodali, A., et al. Atlas: Customizing Large Language Models for Reliable Bibliographic Retrieval and Verification. WASP Workshop, IJCNLP-AACL 2025. aclanthology.org/2025.wasp-main.14

Education

M.S., Computer Science, California State University, Long Beach Aug 2024 - Dec 2026
B.Tech., Computer Science and Engineering, Gandhi Institute of Technology and Management (GITAM) Aug 2020 - May 2024